

1. Complete Technology Stack Matrix

Layer	Technologies Used	Key Purpose & Architectural Function
Frontend Framework	React 19, Vite 8, React Router v7	Ultra-fast SPA rendering, component hierarchy, client routing, optimistic state updates.
Styling System	Tailwind CSS v3, PostCSS, Lucide Icons	Modern glassmorphism, responsive grid layouts, high-contrast accessible civic UI.
Auth & RBAC	Supabase Auth, OAuth 2.0 (Google, GitHub)	Passwordless secure OAuth login, JWT sessions, first-time role onboarding modal, RBAC.
Database & Persistence	Supabase (PostgreSQL 15), Row Level Security	Relational schema (profiles, problems, solutions, projects, messages), real-time sync.
Backend & API Gateway	Vercel Serverless (/api/chat.js), Vite Dev Middleware, Node Express	Zero-exposure environment secrets handling, CORS, rate limiting, and multi-model AI routing.
Artificial Intelligence	Google Gemini API (@google/genai, gemini-2.5-pro, gemini-2.5-flash)	25 strict civic domains, out-of-domain rejection, step-by-step problem-solving guidance.

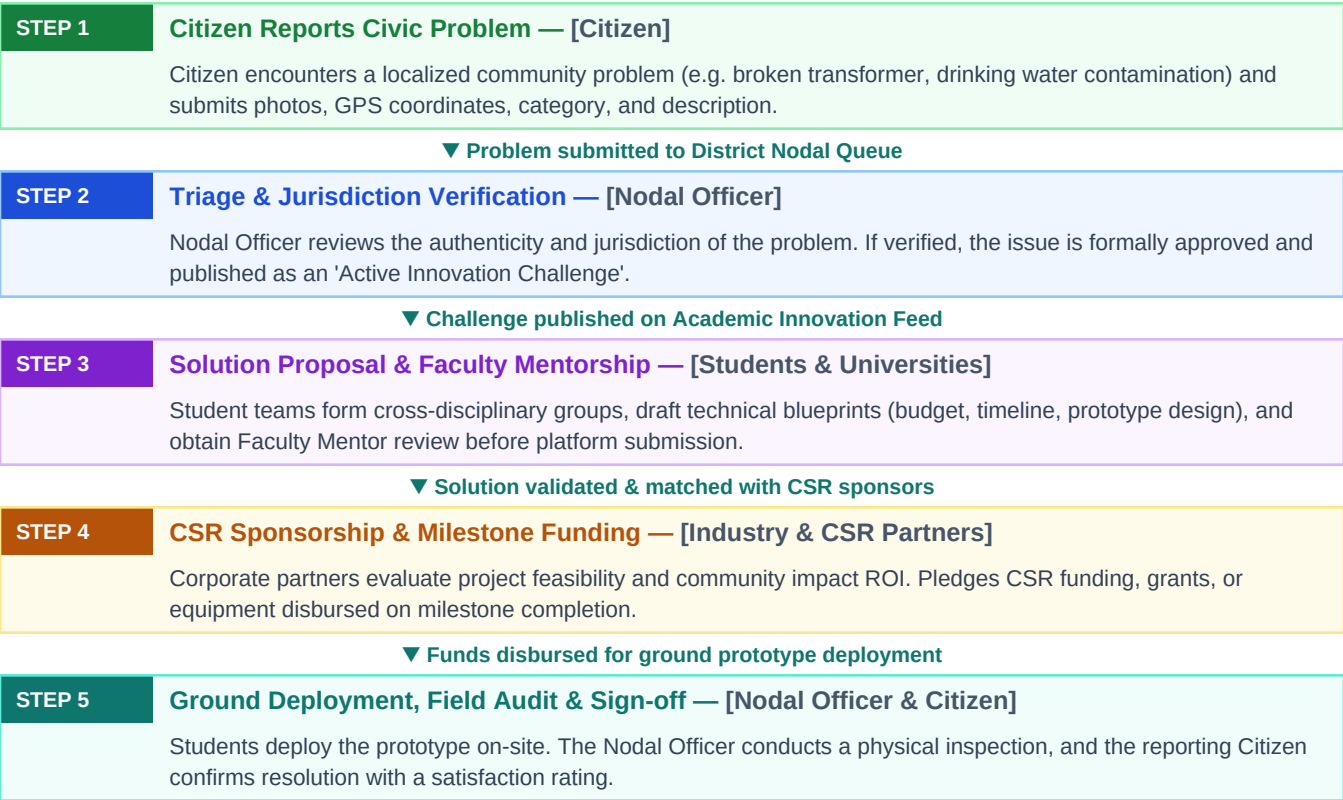
2. How the Backend Works (In Simple Language)

The **Samadhan.Connect Backend** coordinates data flow between the browser, database, and AI engine in 3 core steps:

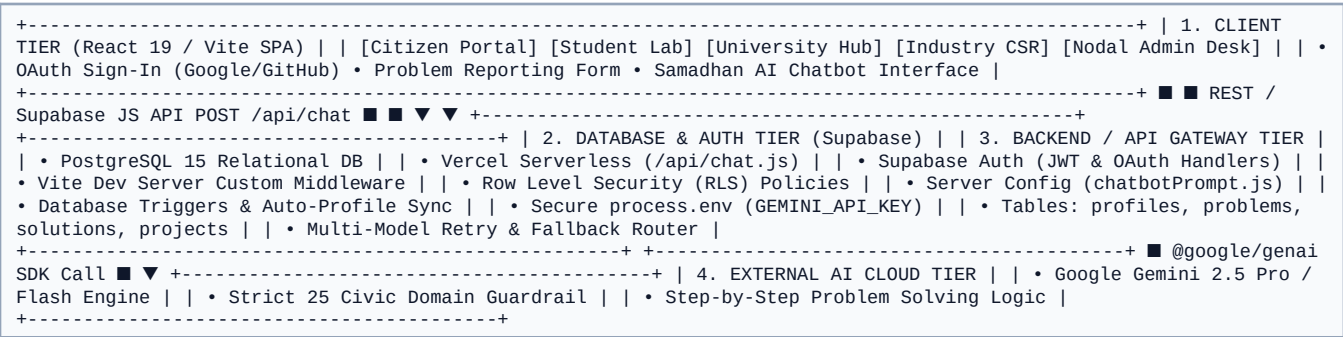
- Secure Gateway & Key Isolation:** When a user asks a question or submits a challenge, the frontend calls POST /api/chat. The backend decrypts server environment variables (like GEMINI_API_KEY), ensuring API keys never touch the browser.
- Dual-Engine Resilience:** The backend attempts live reasoning via Google Gemini 2.5 Pro / Flash. If quota or network limits occur, it seamlessly falls back to the **Intelligent Civic Knowledge Engine**, returning verified step-by-step answers without failing.
- Role-Based Data Security:** Supabase PostgreSQL uses Row Level Security (RLS) and database triggers to guarantee that citizens only edit their own problems, students collaborate securely, and Nodal Officers sign off on genuine civic projects.

3. Flow Chart 1: Master Civic Problem-Solving Lifecycle

This visual flowchart outlines the multi-stakeholder journey from initial citizen problem reporting to CSR funding, deployment, and final verification.



4. Flow Chart 2: System Architecture & Data Pipeline



5. Flow Chart 3: Authentication & Role Onboarding Lifecycle

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[ User clicks 'Sign in with Google' or 'Sign in with GitHub' on Landing Page ] ■ ▼ [ Supabase OAuth handles 3rd-Party Authentication & returns JWT Session Token ] ■ ▼ [ PostgreSQL Database Trigger checks if user exists in 'profiles' table ] ■■■ Existing Profile ■■> [ Load Profile & Redirect to Role Workspace ] ■■■ New Profile ■■■■■■■■■■> [ Insert Basic Profile with Email & Name ] ■ ▼ [ Frontend detects missing role and triggers 'RoleSelectionModal' popup ] [ User selects 1 of 5 Roles: Citizen | Student | University | Industry | Nodal Officer ] ■ ▼ [ DataService updates profile in Supabase DB with selected Role ] ■ ▼ [ Special Condition Check: Is email == 'microsoft1gab@gmail.com'? ] ■■■ YES ■■> [ Grant SUPER_ADMIN role with full /admin panel access to promote users ] ■■■ NO ■■> [ Grant Standard Verified Role Workspace Dashboard ]
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6. Flow Chart 4: Samadhan AI 25-Domain Decision Engine

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[ User enters message or clicks Suggested Domain Chip (e.g. '■ Agriculture', '■ Electricity') ] ■ ▼ [ Frontend appends message to State + Truncates conversation history to last 10 messages ] ■ ▼ [ POST /api/chat payload: { message: "...", history: [...] } ] ■ ▼ [ Backend Gateway checks process.env.GEMINI_API_KEY ] ■■■ API Key Present ■■> [ Initialize @google/genai SDK with SAMADHAN_SYSTEM_PROMPT ] ■ ■ ■ ■ ▼ ■ [ Call Gemini 2.5 Pro / Flash Candidate Models ] ■ ■■■ Success ■■> [ Return JSON reply & status: success ] ■ ■■■ Rate Limit / 404 ■■> [ Failover to Local Knowledge Engine ] ■ ■■■ No API Key ■■■■■■■■■■> [ Run Intelligent Local Knowledge Engine ] ■■■ Is In-Domain? ■■■■■> [ Return Step-by-Step Civic Steps ] ■■■ Is Out-of-Domain? ■■> [ Return Official Canned Refusal ] ■ ▼ [ Frontend receives JSON reply & renders via MarkdownMessage component ] [ Formats Bold text, Bullet points, Numbered steps, and Verification Disclaimers ]
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7. The 25 Authorized Civic Domains

1. ■ Agriculture & Farming • 2. ■ Electricity & Power • 3. ■■ Government Services • 4. ■ Citizen Grievances • 5. ■ Education & Students • 6. ■ Jobs & Employment • 7. ■ Finance & Subsidies • 8. ■ Healthcare & Public Health • 9. ■ Transport & Mobility • 10. ■ Housing & Property • 11. ■■ Weather & Disaster • 12. ■ Environment & Pollution • 13. ■■■■ Legal & Documentation • 14. ■■■■ Business & Startups • 15. ■ Marketplace & SHGs • 16. ■ Digital & Online Services • 17. ■ Identity Documents (Aadhaar/Voter) • 18. ■ Career & Skills • 19. ■ Banking & Insurance • 20. ■■ Municipal & Community Services • 21. ■ Water & Sanitation • 22. ■ Cybersecurity & Online Safety • 23. ■ Tax & Financial Documentation • 24. ■ Emergency & Public Safety • 25. ■ General AI Assistant (Platform Navigation & Problem Reporting).

Specification Summary: This document provides the authoritative technical, operational, and algorithmic workflows for Samadhan.Connect. Engineered for state-wide citizen empowerment, robust security, and seamless civic collaboration.